

with an order that was smaller than your order, otherwise your entire order would have been executed. If you enter an order into a dark pool but do not have any shares traded then there were not any counterparties in the dark pool at that time.

Get-Me-Done. The get-me-done type of algorithms will trade at a specified rate until the order is complete. However, if there is ample liquidity in the order book the algorithm will accelerate trading and sweep the book providing that doing so will complete the order. Usually traders will not want to accelerate trading and sweep the book complete, because by doing so they will likely signal their trading intentions to the market which may result in higher future prices (buy order), lower future prices (sell order), and higher permanent impact for all orders. But in the case where the order would be completed by sweeping the book the less favorable future prices will not affect the performance of the investor. One way to disguise trading intentions is to sweep all liquidity in the book except for the final 100 shares so that you do not affect the market price or NBBO. Many algorithms are set up to react to changing prices or quotes and utilizing this type of sweeping technique would keep those algorithms at bay—at least for the time being anyway.

When we analyze our trading goals it is important to point out that every BES has an expected cost (mean) and timing risk (uncertainty) component. Once the strategy on the efficient trading frontier is determined the expected cost and timing risk value are shown by drawing a horizontal line from the strategy to the y-axis to determine the expected cost and a vertical line from the strategy to the x-axis to determine the corresponding timing risk. Additionally, drawing a line that is tangent to the strategy on the efficient trading frontier to the y-axis results in the value where the strategy will have the highest probability of out-performing. In all case these values are unique except for the situation where we seek to minimize cost (market impact and alpha). In this case, the tangent from strategy A1 (Figure 8.2a) will be a horizontal line, and the expected cost of the strategy and the cost where the strategy will have the highest probability of outperforming will be the same. The strategy will have a 50% chance of outperforming that particular cost.

3) Specify Adaptation Tactic

The third step in the algorithmic decision making process is to specify how the algorithm is to adapt to changing market conditions. This is also commonly referred to as dynamic optimization or adaptive pricing. Mathematicians may cringe at phrase “dynamic optimization” in this